Zewail City of Science, Technology and Innovation

University of Science and Technology

School of Computational Sciences and Artificial Intelligence

CSAI 203 - Fall 2026

Introduction to Software Engineering

Campus Event & Student Club Hub

Software Requirements Specification (SRS)

Team Number: #

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1. Functional Requirements (20 Features)

1. **User Registration:** Students, club managers, and Zewail City admins can create accounts using their university emails, full names, and secure passwords.
2. **Authentication & Login:** The application will allow registered users to log in safely. It will also have an option to reset forgotten passwords via email.
3. **Profile Management:** Users can view and update their profile details, like uploading a profile picture and changing their major.
4. **Club Profile Creation:** The platform must provide admins the ability to create profiles for student clubs, adding the club's name, logo, description, and assigning a manager.
5. **Event Publishing:** Club managers should be able to create events by filling out a form with the event's title, date, time, location, capacity, and description.
6. **Event Modification:** Managers can edit or cancel their events. If an event is canceled, registered students will get an automatic notification.
7. **Event Dashboard:** The home page will display all approved upcoming activities so students can browse Zewail City campus events easily.
8. **Filter and Search:** Users can search for specific events using keywords or filter them by date range and club category.

9. **Event Registration:** Logged-in students can visit an event page and click to register. The system will check if there are empty seats before confirming the booking.
10. **Registration Cancellation:** The app allows students to cancel their booking before the event starts, making the seat available for someone else.
11. **Digital Ticket Generation:** After a successful registration, the system will generate a digital ticket with a QR code for the student.
12. **Attendance Tracking:** Club organizers can scan the students' QR codes at the door to quickly mark them as attended in the database.
13. **Budget Allocation:** Club managers will have a dashboard to see and track their club's available financial budget.
14. **Expense Tracking:** The application lets club managers log costs for specific events, which will be deducted automatically from their budget.
15. **Admin Approval Workflow:** University admins must review submitted events and choose to either approve or reject them before they go live.
16. **Club Subscription:** Students can hit "subscribe" on their favorite student clubs to easily follow their news and activities.
17. **Notifications:** The platform will send alerts to remind students about their upcoming events or when a club they follow posts something new.

18. **Event Feedback:** After an event finishes, attendees can leave a rating and a short text review for the organizing team.
19. **Calendar View:** The system includes a monthly calendar view showing all campus events so students can plan without schedule conflicts.
20. **Reporting:** Managers and admins can download reports that show event attendance numbers and how much money was spent.

2. Non-Functional Requirements (5 Features)

1. **Security:** All user passwords will be encrypted using standard encryption methods. Access to sensitive data and admin controls will be restricted based on user roles.
2. **Performance:** The event dashboard should load in less than 2 seconds, and digital tickets should appear instantly after registration.
3. **Usability:** The user interface will be easy to use and navigate. It must work smoothly and look good on both laptops and mobile phones.
4. **Availability:** The application should be up and running 99.9% of the time so students don't face issues when trying to view their tickets.
5. **Scalability:** The backend is designed to handle up to 1,000 users at the same time without slowing down or crashing, especially when big Zewail City events are announced.